NumPy vs Pandas

Complete color-coded comparison cheatsheet | Readable, wrapped Python examples | Beginner to advanced

Examples are independent snippets. Common aliases: **np** = NumPy, **pd** = Pandas; variables such as a, df, s, x, and rng assume suitable input data. Some operations are conceptually comparable rather than identical.

Foundations

OPERATION	NUMPY	PANDAS
Import	`import numpy as np`	`import pandas as pd`
Create 1D	`np.array([1,2,3])`	`pd.Series([1,2,3])`
Create 2D	`np.array([[1,2],[3,4]])`	`pd.DataFrame({"a":[1,3],"b":[2,4]})`
Range	`np.arange(0,10,2)`	`pd.RangeIndex(0,10,2)`
Even spacing	`np.linspace(0,1,5)`	`pd.Series(np.linspace(0,1,5))`
Zeros	`np.zeros((3,2))`	`pd.DataFrame(0,index=range(3),columns=["a","b"])`
Ones	`np.ones((2,3))`	`pd.DataFrame(1,index=range(2),columns=list("abc"))`
Identity	`np.eye(3)`	`pd.DataFrame(np.eye(3))`
Repeat	`np.repeat([1,2],2)`	`pd.Series([1,2]).repeat(2)`
Tile	`np.tile([1,2],3)`	`pd.Series(np.tile([1,2],3))`
Random generator	`rng=np.random.default_rng(42)`	`pd.DataFrame(rng.normal(size=(5,2)))`
Shape	`a.shape`	`df.shape`

OPERATION	NUMPY	PANDAS
Dimension	`a.ndim`	`df.ndim`
Size	`a.size`	`df.size`
Data types	`a.dtype`	`df.dtypes`
Convert dtype	`a.astype(float)`	`df.astype({"x":"float64"})`
Memory	`a.nbytes`	`df.memory_usage(deep=True).sum()`
Preview	`a[:5]`	`df.head(5)`
Last rows	`a[-5:]`	`df.tail(5)`
Summary	`np.describe not available; np.mean(a)`	`df.describe()`
Metadata	`a.flags`	`df.info()`
Copy	`a.copy()`	`df.copy()`
To Python	`a.tolist()`	`df.to_dict("records")`
Convert across	`df.to_numpy()`	`pd.DataFrame(a)`

Indexing & Filtering

OPERATION	NUMPY	PANDAS
First item	`a[0]`	`s.iloc[0]`
Cell	`a[0,1]`	`df.iloc[0,1]`
Row	`a[0,:]`	`df.iloc[0,:]`

OPERATION	NUMPY	PANDAS
Column	`a[:,0]`	`df["x"]`
Slice	`a[1:4]`	`df.iloc[1:4]`
Label selection	`No labels in ndarray`	`df.loc["r1","x"]`
Multiple columns	`a[:,[0,2]]`	`df[["x","z"]]`
Boolean mask	`a[a>5]`	`df[df["x"]>5]`
AND condition	`a[(a>2)&(a<8)]`	`df[(df.x>2)&(df.x<8)]`
OR condition	`a[(a<2)|(a>8)]`	`df[(df.x<2)|(df.x>8)]`
Membership	`np.isin(a,[2,4])`	`df.x.isin([2,4])`
Where	`np.where(a>0,a,0)`	`df.x.where(df.x>0,0)`
Conditional selection	`np.select([a<0,a==0],[-1,0],default=1)`	`np.select([df.x<0,df.x==0],[-1,0],default=1)`
Filter rows by expression	`a[a[:,0]>10]`	`df.query("x > 10")`
Assign value	`a[0]=99`	`df.loc[0,"x"]=99`
Insert column	`np.column_stack([a,new_col])`	`df["new"]=new_col`
Drop column	`np.delete(a,1,axis=1)`	`df.drop(columns="x")`
Rename columns	`No native column names`	`df.rename(columns={"x":"amount"})`
Reset index	`No index labels`	`df.reset_index(drop=True)`
Set index	`No index labels`	`df.set_index("id")`

Math & Statistics

OPERATION	NUMPY	PANDAS
Add	`a+b`	`df1.add(df2,fill_value=0)`
Subtract	`a-b`	`df1.sub(df2)`
Multiply	`a*b`	`df1.mul(df2)`
Divide	`a/b`	`df1.div(df2)`
Power	`a**2`	`df.pow(2)`
Modulo	`a%2`	`df.mod(2)`
Floor divide	`a//2`	`df.floordiv(2)`
Absolute	`np.abs(a)`	`df.abs()`
Round	`np.round(a,2)`	`df.round(2)`
Square root	`np.sqrt(a)`	`np.sqrt(df)`
Exponential	`np.exp(a)`	`np.exp(df)`
Natural log	`np.log(a)`	`np.log(df)`
Trig	`np.sin(a)`	`np.sin(df)`
Clip	`np.clip(a,0,100)`	`df.clip(0,100)`
Minimum	`np.min(a,axis=0)`	`df.min(axis=0)`
Maximum	`np.max(a,axis=0)`	`df.max(axis=0)`

OPERATION	NUMPY	PANDAS
Sum	`np.sum(a,axis=0)`	`df.sum(axis=0)`
Mean	`np.mean(a,axis=0)`	`df.mean(axis=0)`
Median	`np.median(a,axis=0)`	`df.median(axis=0)`
Std dev (sample)	`np.std(a,ddof=1)`	`df.std(ddof=1)`
Variance (sample)	`np.var(a,ddof=1)`	`df.var(ddof=1)`
Percentile	`np.percentile(a,95)`	`df.quantile(.95)`
Cumulative sum	`np.cumsum(a)`	`df.cumsum()`
Cumulative product	`np.cumprod(a)`	`df.cumprod()`
Difference	`np.diff(a)`	`df.diff()`
Product	`np.prod(a)`	`df.prod()`
Count nonzero	`np.count_nonzero(a)`	`(df!=0).sum()`
Correlation	`np.corrcoef(x,y)[0,1]`	`df[["x","y"]].corr()`
Covariance	`np.cov(x,y)`	`df[["x","y"]].cov()`
Histogram bins	`np.histogram(a,bins=10)`	`df.x.plot.hist(bins=10)`
Weighted average	`np.average(a,weights=w)`	`np.average(df.x,weights=df.w)`
Unique	`np.unique(a)`	`df.x.unique()`
Counts	`np.unique(a,return_counts=True)`	`df.x.value_counts()`

OPERATION	NUMPY	PANDAS
All / any	`np.all(a>0); np.any(a>0)`	`(df.x>0).all(); (df.x>0).any()`

Missing & Cleaning

OPERATION	NUMPY	PANDAS
Detect NaN	`np.isnan(a)`	`df.isna()`
Detect finite	`np.isfinite(a)`	`np.isfinite(df.select_dtypes("number"))`
Detect infinity	`np.isinf(a)`	`np.isinf(df.select_dtypes("number"))`
NaN-safe sum	`np.nansum(a)`	`df.sum()`
NaN-safe mean	`np.nanmean(a)`	`df.mean()`
Replace NaN	`np.nan_to_num(a,nan=0)`	`df.fillna(0)`
Mean imputation	`np.where(np.isnan(a),np.nanmean(a),a)`	`df.x.fillna(df.x.mean())`
Remove missing	`a[~np.isnan(a)]`	`df.dropna()`
Forward fill	`No direct ndarray equivalent`	`df.ffill()`
Backward fill	`No direct ndarray equivalent`	`df.bfill()`
Interpolate	`np.interp(idx,known_idx,known_vals)`	`df.interpolate()`
Duplicates	`np.unique(a,axis=0)`	`df.drop_duplicates()`
Duplicate mask	`No direct row mask helper`	`df.duplicated()`
Replace values	`np.where(a==0,-1,a)`	`df.replace(0,-1)`

OPERATION	NUMPY	PANDAS
Outlier IQR	`q1,q3=np.percentile(a,[25,75])`	`q1,q3=df.x.quantile([.25,.75])`
Z score	`(a-a.mean())/a.std()`	`(df.x-df.x.mean())/df.x.std(ddof=0)`
Numeric coercion	`a.astype(float) # invalid raises`	`pd.to_numeric(df.x,errors="coerce")`
Null count	`np.isnan(a).sum()`	`df.isna().sum()`

Reshape & Combine

OPERATION	NUMPY	PANDAS
Reshape	`a.reshape(2,3)`	`df.to_numpy().reshape(2,3)`
Transpose	`a.T`	`df.T`
Flatten	`a.ravel()`	`df.to_numpy().ravel()`
Stack vertically	`np.vstack([a,b])`	`pd.concat([df1,df2],ignore_index=True)`
Stack horizontally	`np.hstack([a,b])`	`pd.concat([df1,df2],axis=1)`
Concatenate	`np.concatenate([a,b],axis=0)`	`pd.concat([df1,df2])`
Split	`np.array_split(a,3)`	`np.array_split(df,3) # positional`
Insert	`np.insert(a,1,9)`	`pd.concat([df.iloc[:1],new,df.iloc[1:]])`
Delete	`np.delete(a,1,axis=0)`	`df.drop(index=1)`
Swap axes	`np.swapaxes(a,0,1)`	`df.T # 2D only`
Broadcast	`a+np.array([1,2,3])`	`df.add(pd.Series([1,2,3],index=df.columns))`

OPERATION	NUMPY	PANDAS
Pivot	`No label-aware pivot`	`df.pivot(index="id",columns="kind",values="v")`
Pivot table	`No built-in grouped pivot`	`df.pivot_table(index="kind",values="v",aggfunc="mean")`
Melt	`a.reshape(-1) # loses labels`	`df.melt(id_vars="id")`
Crosstab	`np.histogram2d(x,y)`	`pd.crosstab(df.a,df.b)`
One-hot encode	`np.eye(k)[labels]`	`pd.get_dummies(df["category"])`
SQL join	`No native key-based join`	`df1.merge(df2,on="id",how="left")`
Inner join	`No native key-based join`	`df1.merge(df2,on="id",how="inner")`
Outer join	`No native key-based join`	`df1.merge(df2,on="id",how="outer")`
Index join	`No index labels`	`df1.join(df2,how="left")`
Join validation	`No native validation`	`df1.merge(df2,on="id",validate="one_to_one")`

Sort, Search & Groups

OPERATION	NUMPY	PANDAS
Sort	`np.sort(a)`	`df.sort_values("x")`
Sort indices	`np.argsort(a)`	`df.x.argsort()`
Descending	`np.sort(a)[::-1]`	`df.sort_values("x",ascending=False)`
Search sorted	`np.searchsorted(a_sorted,5)`	`df.x.searchsorted(5)`
Find positions	`np.where(a>5)[0]`	`df.index[df.x>5]`

OPERATION	NUMPY	PANDAS
Arg min/max	`np.argmin(a); np.argmax(a)`	`df.x.idxmin(); df.x.idxmax()`
Top N	`np.sort(a)[-3:]`	`df.nlargest(3,"x")`
Rank	`np.argsort(np.argsort(a))`	`df.x.rank()`
Set union	`np.union1d(a,b)`	`pd.Index(a).union(pd.Index(b))`
Intersection	`np.intersect1d(a,b)`	`pd.Index(a).intersection(pd.Index(b))`
Difference	`np.setdiff1d(a,b)`	`pd.Index(a).difference(pd.Index(b))`
Group by	`np.unique(keys,return_inverse=True)`	`df.groupby("category")["x"].mean()`
Group aggregate	`np.bincount(groups,weights=values)`	`df.groupby("category").agg({"x":["sum","mean"]})`
Group transform	`Manual group mapping`	`df.groupby("category").x.transform("mean")`
Group filter	`Manual mask by group`	`df.groupby("category").filter(lambda g:len(g)>2)`
Frequency	`np.unique(a,return_counts=True)`	`df.x.value_counts(normalize=True)`

Time, Text & Windows

OPERATION	NUMPY	PANDAS
String lower	`np.char.lower(a.astype(str))`	`df.name.str.lower()`
String contains	`np.char.find(a.astype(str),"QA")>=0`	`df.name.str.contains("QA",na=False)`
String strip	`np.char.strip(a.astype(str))`	`df.name.str.strip()`
Regex extract	`No native regex extraction`	`df.name.str.extract(r"(\d+)")`

OPERATION	NUMPY	PANDAS
Split text	`np.char.split(a.astype(str),",")`	`df.name.str.split(",")`
Date conversion	`a.astype("datetime64[D]")`	`pd.to_datetime(df.date)`
Year	`a.astype("datetime64[Y]")`	`df.date.dt.year`
Time delta	`np.diff(dates)`	`df.date.diff()`
Resample time	`Manual binning`	`df.set_index("date").resample("D").sum()`
Shift / lag	`np.roll(a,1) # wraps!`	`df.x.shift(1) # missing first`
Rolling mean	`np.convolve(a,np.ones(3)/3,mode="valid")`	`df.x.rolling(3).mean()`
Expanding mean	`np.cumsum(a)/np.arange(1,len(a)+1)`	`df.x.expanding().mean()`
EWMA	`Manual recurrence`	`df.x.ewm(span=5,adjust=False).mean()`
Categorical dtype	`No category metadata`	`df.x.astype("category")`
MultiIndex	`No native hierarchical index`	`df.set_index(["country","city"])`
Matrix multiply	`a@b`	`df1@df2`
Dot product	`np.dot(a,b)`	`df1.dot(df2)`
Inverse	`np.linalg.inv(a)`	`pd.DataFrame(np.linalg.inv(df))`
Determinant	`np.linalg.det(a)`	`np.linalg.det(df)`
Eigenvalues	`np.linalg.eig(a)`	`np.linalg.eig(df)`
SVD	`np.linalg.svd(a)`	`np.linalg.svd(df)`

OPERATION	NUMPY	PANDAS
Solve linear system	`np.linalg.solve(A,b)`	`np.linalg.solve(df_A,s_b)`
Norm	`np.linalg.norm(a)`	`np.linalg.norm(df.to_numpy())`
Random uniform	`rng.uniform(0,1,5)`	`pd.Series(rng.uniform(0,1,5))`
Random normal	`rng.normal(0,1,5)`	`pd.Series(rng.normal(0,1,5))`
Random integers	`rng.integers(0,10,5)`	`pd.Series(rng.integers(0,10,5))`
Shuffle	`rng.permutation(a)`	`df.sample(frac=1,random_state=42)`
Sample rows	`rng.choice(a,3,replace=False)`	`df.sample(n=3,random_state=42)`
Seed reproducibly	`rng=np.random.default_rng(42)`	`df.sample(frac=.2,random_state=42)`

I/O, Plots & Performance

OPERATION	NUMPY	PANDAS
Read CSV	`np.loadtxt("data.csv",delimiter=",",skiprows=1)`	`pd.read_csv("data.csv")`
Save CSV	`np.savetxt("out.csv",a,delimiter=",")`	`df.to_csv("out.csv",index=False)`
Load binary	`np.load("arr.npy")`	`pd.read_pickle("frame.pkl")`
Save binary	`np.save("arr.npy",a)`	`df.to_pickle("frame.pkl")`
Compressed arrays	`np.savez_compressed("a.npz",a=a)`	`df.to_parquet("frame.parquet")`
Excel	`No built-in Excel reader`	`pd.read_excel("data.xlsx")`
JSON	`No built-in JSON DataFrame parser`	`pd.read_json("data.json")`

OPERATION	NUMPY	PANDAS
SQL	`No native SQL client`	`pd.read_sql(query,connection)`
Parquet	`No built-in Parquet reader`	`pd.read_parquet("data.parquet")`
Read chunks	`Manual file chunks`	`pd.read_csv("data.csv",chunksize=10000)`
Line chart	`plt.plot(a)`	`df.x.plot.line()`
Bar chart	`plt.bar(range(len(a)),a)`	`df.x.plot.bar()`
Histogram	`plt.hist(a,bins=20)`	`df.x.plot.hist(bins=20)`
Scatter	`plt.scatter(x,y)`	`df.plot.scatter(x="x",y="y")`
Boxplot	`plt.boxplot(a)`	`df.boxplot(column="x")`
Vectorize	`a*2 # native ufuncs`	`df.x*2 # vectorized arithmetic`
Custom apply	`np.vectorize(f)(a) # not faster`	`df.x.map(f) # Python-level`
Memory reduction	`a.astype(np.float32)`	`df.x.astype("float32")`
Chunk computation	`np.array_split(a,10)`	`pd.read_csv("data.csv",chunksize=10000)`
Method chaining	`f(g(a))`	`df.dropna().assign(z=lambda d:d.x*2)`
Pipe	`f(a)`	`df.pipe(f)`
Style output	`No native HTML styler`	`df.style.format({"x":"{:.2f}"})`
Copy-on-write	`Views may share memory`	`pd.options.mode.copy_on_write=True`

Advanced & Ai/Ml Qa

OPERATION	NUMPY	PANDAS
Einsum	`np.einsum("ij,jk->ik",A,B)`	`np.einsum("ij,jk->ik",df1,df2)`
Masked array	`np.ma.masked_invalid(a)`	`df.mask(df<0)`
Structured array	`np.array([(1,2.)],dtype=[("id","i4"),("x","f8")])`	`pd.DataFrame({"id":[1],"x":[2.]})`
Sliding windows	`np.lib.stride_tricks.sliding_window_view(a,3)`	`df.x.rolling(3)`
Unique rows	`np.unique(a,axis=0)`	`df.drop_duplicates()`
Data validation	`assert np.isfinite(a).all()`	`assert df["x"].notna().all()`
Range checks	`assert ((a>=0)&(a<=1)).all()`	`assert df.x.between(0,1).all()`
Schema check	`assert a.ndim==2`	`assert set(["id","x"]).issubset(df.columns)`
Row count	`assert len(a)==expected`	`assert len(df)==expected`
No duplicate IDs	`assert len(np.unique(ids))==len(ids)`	`assert df.id.is_unique`
Feature scaling	`(a-a.mean(0))/a.std(0)`	`(df[cols]-df[cols].mean())/df[cols].std(ddof=0)`
Train/test indices	`rng.permutation(len(a))`	`df.sample(frac=1,random_state=42)`
Drift PSI inputs	`np.histogram(base,bins=edges)[0]`	`pd.cut(df.x,edges).value_counts(sort=False)`
Prediction accuracy	`np.mean(y_true==y_pred)`	`(df.y_true==df.y_pred).mean()`
Confusion counts	`np.histogram2d(y_true,y_pred,bins=[edges,edges])`	`pd.crosstab(df.y_true,df.y_pred)`
Missingness rate	`np.isnan(a).mean(axis=0)`	`df.isna().mean()`
Outlier flags	`np.abs((a-a.mean())/a.std())>3`	`(df.x-df.x.mean()).abs()>3*df.x.std()`

OPERATION	NUMPY	PANDAS
Data parity	`np.allclose(a,b,equal_nan=True)`	`pd.testing.assert_frame_equal(df1,df2)`
Schema parity	`a.shape==b.shape`	`df1.dtypes.equals(df2.dtypes)`
Floating tolerance	`np.isclose(a,b,rtol=1e-5)`	`np.isclose(df1.x,df2.x,rtol=1e-5)`
Group fairness rate	`Manual group masks`	`df.groupby("group").pred.mean()`
Model score distribution	`np.histogram(scores,bins=10)`	`df.score.describe()`